

UNIT 3: Water Chemistry & Phase Rule

1. Hardness of Water & EDTA Method

Hardness is the property of water that prevents lather formation with soap due to dissolved Ca^{2+} and Mg^{2+} salts.

Types of Hardness:

- Temporary Hardness (Carbonate): Caused by bicarbonates $\text{Ca}(\text{HCO}_3)_2$ & $\text{Mg}(\text{HCO}_3)_2$. Easily removed by boiling.
- Permanent Hardness (Non-Carbonate): Caused by chlorides & sulfates CaCl_2 , MgSO_4 . Cannot be removed by boiling.
- Units Conversion: $1 \text{ ppm} = 1 \text{ mg/L} = 0.07^\circ\text{Cl} = 0.1^\circ\text{Fr}$.

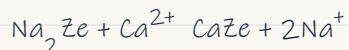
EDTA Titration Method:

- Indicator: Eriochrome Black T (EBT) at pH 10 (Ammonia buffer $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$).
- Color Change: Wine Red (Ca/Mg-EBT complex) Steel Blue (Free EBT at endpoint). EDTA forms 1:1 stable complex with $\text{Ca}^{2+}/\text{Mg}^{2+}$.

2. Water Softening Processes

A. Zeolite (Permutit) Process:

Hydrated sodium aluminosilicate $\text{Na}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot x\text{SiO}_2 \cdot y\text{H}_2\text{O}$ (Na_2Ze) exchanges Na^+ for $\text{Ca}^{2+}/\text{Mg}^{2+}$.



Regeneration: Washed with 10% NaCl brine solution: $\text{CaZe} + 2\text{NaCl} \rightarrow \text{Na}_2\text{Ze} + \text{CaCl}_2$.

B. Ion-Exchange Process (Demineralization):

- Cation Exchanger (RH^+): Exchanges Ca^{2+} , Mg^{2+} for H^+ .
- Anion Exchanger (ROH^-): Exchanges Cl^- , SO_4^{2-} for OH^- .
- $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$. Produces ultra-pure demineralized water for high-pressure boilers!

3. Boiler Troubles

1. Scale vs Sludge: Sludge is loose, slimy precipitate formed in cooler boiler zones; Scale is hard, adherent coating on hot inner walls causing overheating.
2. Priming & Foaming: Priming = carryover of water droplets in steam; Foaming = production of persistent bubbles.
3. Caustic Embrittlement: Cracking of boiler metal due to high NaOH concentration in hairline stress cracks.

4. Phase Rule & Water System

Gibbs Phase Rule Equation:

$$F = C - P + 2$$

where F = Degrees of Freedom, C = Components, P = Phases.

Single Component Water System ($C = 1$):

- Phases: Ice (solid), Water (liquid), Vapour (gas).
- Curves (OA, OB, OC): 2 phases in equilibrium ($P = 2$). $F = 1 - 2 + 2 = 1$ (Univariant).
- Areas between curves: Single phase exists ($P = 1$). $F = 1 - 1 + 2 = 2$ (Bivariant).
- Triple Point 'O': All 3 phases co-exist in equilibrium at $T = 0.0075^\circ\text{C}$, $P = 4.58 \text{ mm Hg}$. $P = 3$, $F = 1 - 3 + 2 = 0$ (Non-variant / Invariant!).

5. Quick Exam Revision Summary

- EDTA Endpoint: Wine red Steel blue.
- Zeolite Regeneration: 10% NaCl solution.
- Gibbs Phase Rule: $F = C - P + 2$.
- Triple Point of Water: $T = 0.0075^\circ\text{C}$, $P = 4.58 \text{ mmHg}$ ($F = 0$).